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Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle MATHS TUTORIALS

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Half Past Paper – Week 15

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
1a	Prime factor decomposition	2	U739
1b	Lowest common multiple, Finding the HCF and LCM using prime factors	2	U751,U250
3	Sharing amounts in a given ratio, Writing numbers as percentages of other numbers	5	U577,U925
5	Finding unknown sides in right-angled triangles	2	U283
7	Using Pythagoras' theorem in 2D, Finding the area of sectors	4	U385,U373
9	Interpreting box plots	2	U837
11	Using the product rule for counting	2	U369
13	The sine rule	3	U952
15	Enlargement by a positive or negative scale factor	2	U134
17	Finding the turning point of a quadratic graph by completing the square	1	U769
19	Substituting into iterative formulae	4	U434
21a	Transforming graphs	2	U455
21b	Transfoming graphs	2	U455
	Total	33	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 (*a*) Write 84 as a product of its prime factors.

.....

(2)

(*b*) Find the lowest common multiple (LCM) of 60 and 84

.....

(2)

(Total for Question 1 is 4 marks)

3 Carlo puts tins into small boxes and into large boxes.

He puts 6 tins into each small box.
He puts 20 tins into each large box.

Carlo puts a total of 3000 tins into the boxes so that

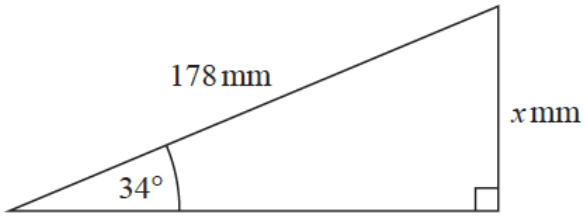
number of tins in small boxes : number of tins in large boxes = 2 : 3

Carlo says that less than 30% of the boxes filled with tins are large boxes.

Is Carlo correct?
You must show all your working.

(Total for Question 3 is 5 marks)

5

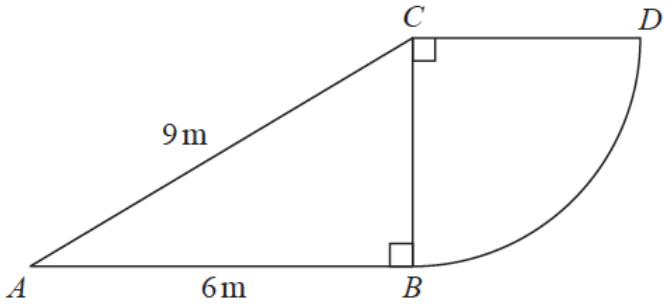


Work out the value of x.
Give your answer correct to 1 decimal place.

.....

(Total for Question 5 is 2 marks)

7 The diagram shows a right-angled triangle and a quarter circle.



The right-angled triangle ABC has angle $ABC = 90°$
The quarter circle has centre C and radius CB.

Work out the area of the quarter circle.
Give your answer correct to 3 significant figures.
You must show all your working.

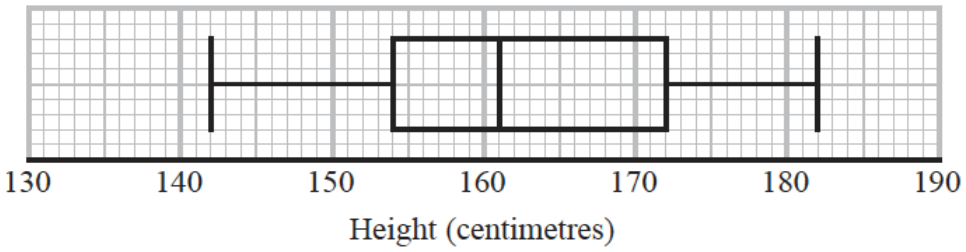
..... m^2

(Total for Question 7 is 4 marks)

9 Aisha recorded the heights, in centimetres, of some girls.
She used her results to work out the information in this table.

Least height	142 cm
Lower quartile	154 cm
Interquartile range	17 cm
Median	162 cm
Range	40 cm

Aisha drew this box plot for the information in the table.
The box plot is **not** fully correct.



Write down the two things Aisha should do to make the box plot fully correct.

1.....

.....

.....

2.....

.....

.....

(Total for Question 9 is 2 marks)

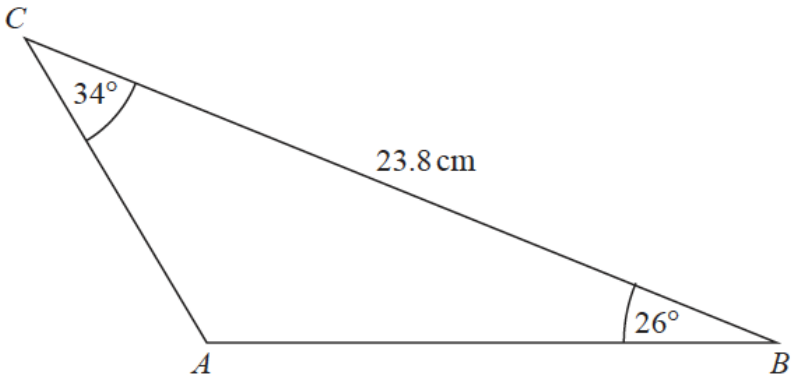
11 Jack is in a restaurant.
There are 5 starters, 8 main courses and some desserts on the menu.

Jack is going to choose one starter, one main course and one dessert.
He says there are 240 ways that he can choose his starter, his main course and his dessert.

Could Jack be correct?
You must show how you get your answer.

(Total for Question 11 is 2 marks)

13 Here is triangle ABC.

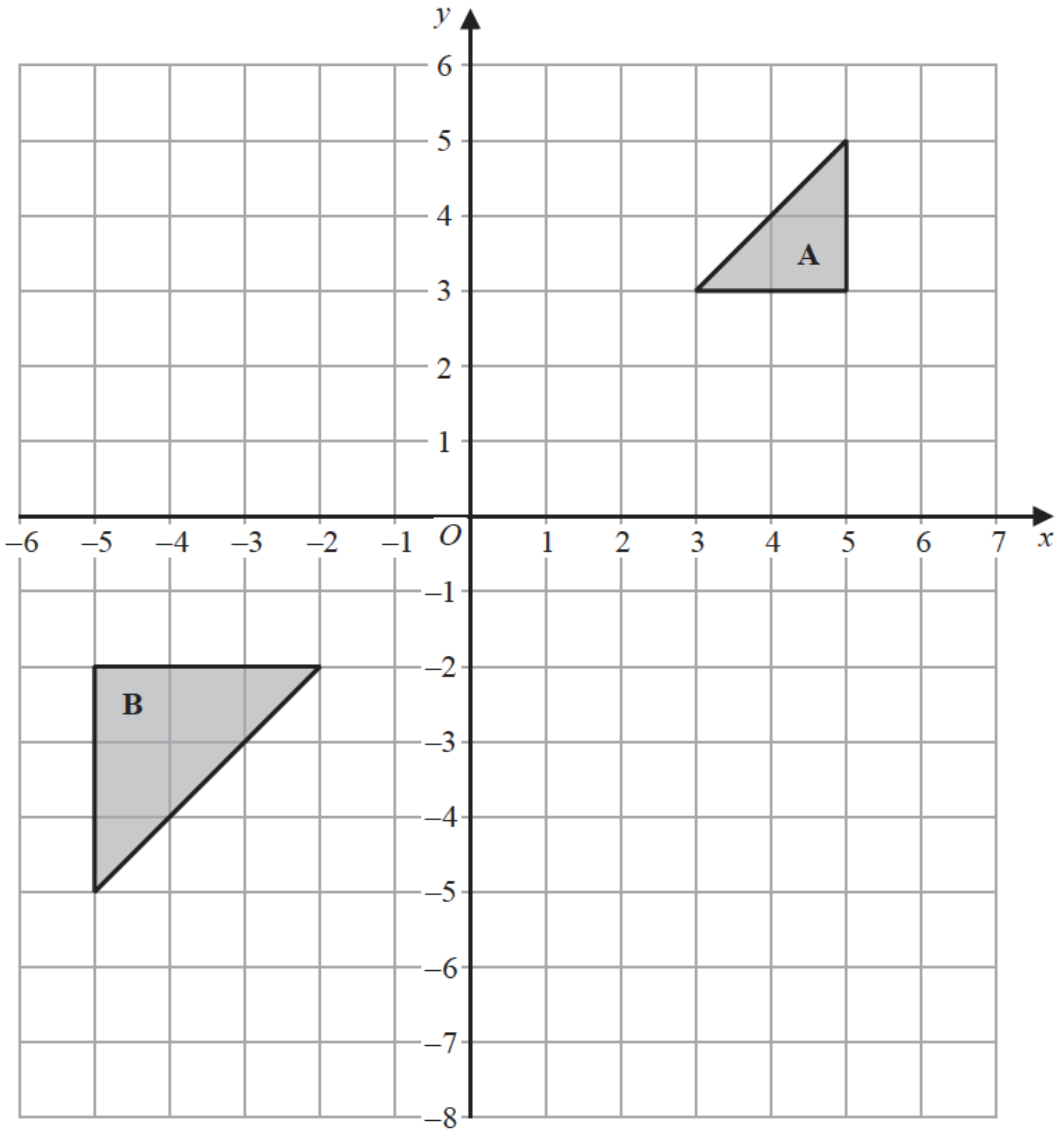


Work out the length of AB.
Give your answer correct to 1 decimal place.

..... cm

(Total for Question 13 is 3 marks)

15



Describe fully the single transformation that maps triangle **A** onto triangle **B**.

.....

.....

(Total for Question 15 is 2 marks)

17 Write down the coordinates of the turning point on the graph of $y = (x + 12)^2 - 7$

(..... , ...)

(Total for Question 17 is 1 mark)

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19 A hot air balloon is descending.
The height of the balloon n minutes after it starts to descend is h_n metres.

The height of the balloon $(n + 1)$ minutes after it starts to descend, h_{n+1} metres, is given by

$$h_{n+1} = K \times h_n + 20 \qquad \text{where } K \text{ is a constant.}$$

The balloon starts to descend from a height of 1200 metres at 09 15
At 09 16 the height of the balloon is 1040 metres.

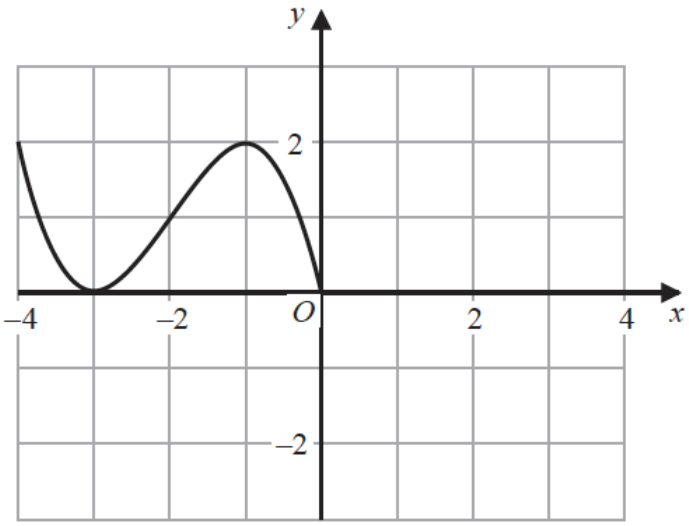
Work out the height of the balloon at 09 18

..... m

(Total for Question 19 is 4 marks)

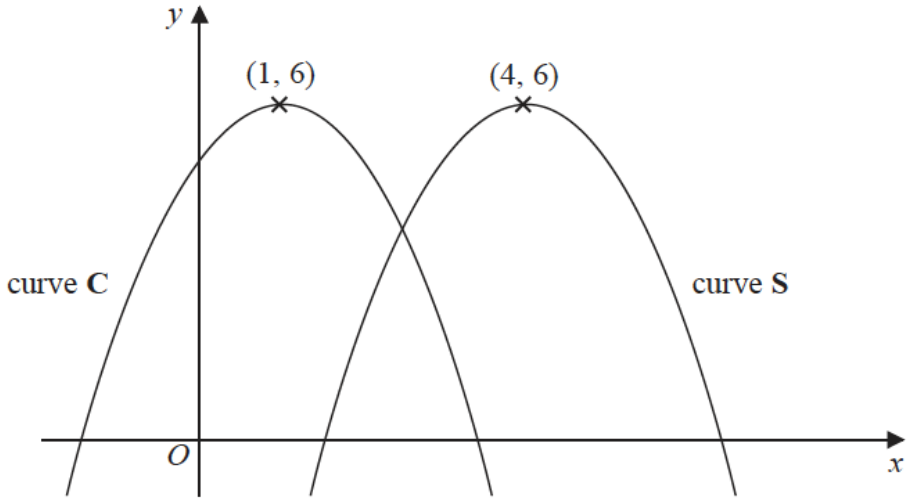
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21 The graph of the curve with equation $y = f(x)$ is shown on the grid below.



(*a*) On the grid above, sketch the graph of the curve with equation $y = f(-x)$

(2)



The curve **C** with equation $y = 5 + 2x - x^2$ is transformed by a translation to give the curve **S** such that the point (1, 6) on **C** is mapped to the point (4, 6) on **S**.

(*b*) Find an equation for **S**.

.....

(2)

(Total for Question 21 is 4 marks)

TOTAL FOR PAPER IS 33 MARKS